

GEOMETRY - HANDOUT 01_EXPLANATORY NOTES

Solutions for questions 1 to 15:

1. (i) Let the angle be θ

Complement of the angle is $\left(\frac{7}{2}\right)$ th of its measure.

$$90^\circ - \theta = \frac{7}{2} \theta$$

$$\theta = 20^\circ$$

Supplementary angle of the angle = $180^\circ - \theta = 160^\circ$

Ans: (160°)

- (ii) Let the angle be θ

$90^\circ - \theta$ and $180^\circ - \theta$ are in the ratio 1 : 3

$$\frac{90^\circ - \theta}{180^\circ - \theta} = \frac{1}{3}$$

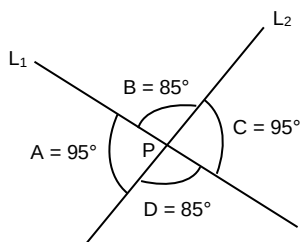
$$270^\circ - 3\theta = 180^\circ - \theta$$

$$90^\circ = 2\theta$$

$$\theta = 45^\circ$$

Ans: (45°)

2. (i)



$$\angle A + \angle D = 180^\circ$$

$$\angle A + 85^\circ = 180^\circ$$

$$\angle A = 95^\circ$$

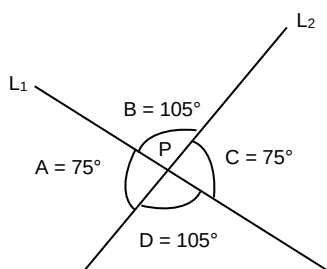
$$\angle B = \angle D = 85^\circ (\because \text{Vertically opposite angles are equal})$$

$$\angle C = \angle A = 95^\circ (\because \text{Vertically opposite angles are equal})$$

$\angle A$ and $\angle C$ are obtuse angles. $\angle B$ and $\angle D$ are acute angles.

Ans: $(95^\circ, 85^\circ, 95^\circ)$

- (ii)



$$\angle A + \angle D = 180^\circ$$

$$\angle A + 105^\circ = 180^\circ$$

$$\angle A = 75^\circ$$

$$\angle B = \angle D = 105^\circ (\because \text{Vertically opposite angles are equal})$$

$$\angle C = \angle A = 75^\circ (\because \text{Vertically opposite angles are equal})$$

$\angle A$ and $\angle C$ are acute angles. $\angle B$ and $\angle D$ are obtuse angles

Ans: $(75^\circ, 105^\circ, 75^\circ)$

- (iii) $\angle A + \angle D = 180^\circ$

$$\angle A + \text{an acute angle} = 180^\circ$$

$$\angle A = 180^\circ - (\text{an acute angle}) = \text{an obtuse angle}$$

Vertically opposite angles are equal

$$\therefore \angle B = \angle D \text{ and } \angle C = \angle A$$

$\angle A$ and $\angle C$ are obtuse angles. $\angle B$ and $\angle D$ are acute angles. The measures of $\angle A$, $\angle B$ and $\angle C$ are obtuse, acute and obtuse respectively.

Ans: (obtuse, acute, obtuse)

3. (i) $7\angle 2 = 11\angle 4$ i.e., $\frac{\angle 2}{\angle 4} = \frac{11}{7}$

$$\text{Also } \angle 2 + \angle 4 = 180^\circ$$

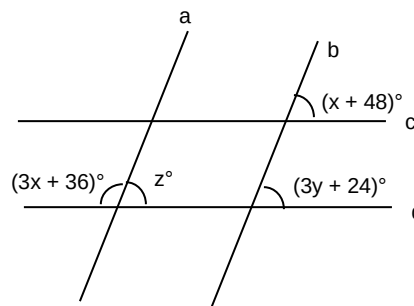
$$\therefore \angle 2 = 110^\circ \text{ and } \angle 4 = 70^\circ$$

$$\angle 7 = \angle 2 (\because \text{Alternate exterior angles are equal})$$

$$\angle 7 = 110^\circ$$

Ans: (110°)

- (ii)



$(x + 48)^\circ = (3y + 24)^\circ (\because c \parallel d)$. So, corresponding angles are equal.

$z^\circ = (3y + 24)^\circ (\because a \parallel b)$. So, corresponding angles are equal

$$(3x + 36)^\circ + z^\circ = 180^\circ$$

$$(3x + 36)^\circ + (3y + 24)^\circ = 180^\circ$$

$$x + y = 40^\circ - (1)$$

$$\text{Also, we got } (x + 48)^\circ = (3y + 24)^\circ$$

$$x + 48^\circ = 3y + 24^\circ$$

$$40^\circ - y + 48^\circ = 3y + 24^\circ (\because \text{From (1)})$$

$$y = 16^\circ$$

Ans: (16°)

4. PQ, RS and TU are parallel lines.

AB and CD are the transversals cutting these parallel lines.

The intercepts formed by the parallel lines are proportional

$$\text{i.e., } \frac{EF}{FG} = \frac{HI}{IJ}$$

$$\frac{1.65}{EG - EF} = \frac{HI}{HJ - HI}$$

$$\frac{1.65}{6.6 - 1.65} = \frac{HI}{6.4 - HI}$$

$$\frac{1}{3} = \frac{HI}{6.4 - HI}$$

$$HI = 1.6$$

$$IJ = 6.4 - HI = 4.8 \text{ cm}$$

Ans: (4.8)

5. (i) $\angle C = 180^\circ - (\text{External angle at C}) = 180^\circ - 130^\circ = 50^\circ$

The triangle is isosceles. Either $\angle C = (\angle A \text{ or } \angle B)$ OR

$$\angle A = \angle B$$

If $\angle C = (\angle A \text{ or } \angle B)$, $(\angle B \text{ or } \angle A)$ will be $180^\circ - 2(50^\circ)$ i.e., 80°

If $\angle A = \angle B$, each of them will be $\frac{180^\circ - \angle C}{2}$ i.e., 65°

The angles of ABC are $50^\circ, 50^\circ, 80^\circ$ or $65^\circ, 65^\circ, 50^\circ$

The largest possible measure of any of the angles of the triangle is 80°

Ans: (80°)

- (ii) $\angle DAB + \angle EBC = 250^\circ$

$$180^\circ - \angle A + 180^\circ - \angle B = 250^\circ$$

$$\angle A + \angle B = 110^\circ \text{ — (1)}$$

$$\angle EBC + \angle BCF = 230^\circ$$

$$180^\circ - \angle B + 180^\circ - \angle C = 230^\circ$$

$$\angle B + \angle C = 130^\circ \text{ --- (2)}$$

$$(1) + (2): \angle B + \angle A + \angle B + \angle C = 240^\circ$$

$$\angle B + 180^\circ = 240^\circ$$

$$\angle B = 60^\circ$$

$$\angle ABC = 60^\circ \quad \text{Ans: } (60^\circ)$$

6. One of the angles of the triangle is 50° . The sum of its other two angles is 130° .

(i) The triangle is acute angled i.e., all the angles of the triangles are acute.

All the angles of the triangle (in degrees) are integers
Let the other two angles of the triangle be a and b
 $a + b = 130^\circ$

(a, b) can be (89, 41), (88, 42), (87, 43), (65, 65)

(a, b) has 25 possible values

Number of combinations of the angles of the triangle is 25.

($\therefore$ Here (a, b) is same as (b, a). Ans: (25)

(ii) Let the other two angles of the triangle be a and b
 $a + b = 130^\circ$

All the angles of the triangle (in degrees) are integers.
The triangle is scalene i.e., no two angles of the triangle are equal.

(a, b) cannot be (50, 80), or (65, 65)

Here (a, b) is same as (b, a)

(a, b) can be (1, 129), (2, 128), (3, 127), (49, 81), (51, 79), (64, 66).

(a, b) has 63 possible values.

Number of combinations of the angles of the triangle is 63.
Ans: (63)

7. Triangle PQR is isosceles.

If $PQ = PR$, $\angle PQR = \angle PRQ$ i.e., $\angle PRQ = 30^\circ$. But $\angle PRQ = \angle RPS + \angle PSQ = \angle RPS + 75^\circ$

$\therefore \angle PRQ > 75^\circ$. This is not possible

If $PQ = QR$, $\angle QPR = \angle PRQ = \frac{180^\circ - \angle PQR}{2} = 75^\circ$. But

$\angle PRQ = \angle RPS + \angle PSQ$

$= \angle RPS + 75^\circ$.

$\therefore \angle PRQ > 75^\circ$.

This is not possible

Only possibility is $PR = QR$

$\therefore \angle QPR = \angle PQR = 30^\circ$

$\angle QPS = 180^\circ - (\angle PQS + \angle PSQ) = 180^\circ - (30^\circ + 75^\circ)$

$= 75^\circ$
 $\angle RPS = \angle QPS - \angle QPR = 75^\circ - 30^\circ = 45^\circ$ Ans: (45°)

8. (i) $7^2 + 24^2 = 25^2$ ($\therefore 7^2 = 49, 24^2 = 576, 25^2 = 625$)
The triangle is right angled. Ans: (right angled)

(ii) $9^2 = 81, 10^2 = 100, 18^2 = 324$
 $9^2 + 10^2 = 181, 9^2 + 10^2 < 18^2$

The triangle is obtuse angled.

Ans: (obtuse angled)

(iii) $8^2 + 8^2 = 8^2(2) = (8\sqrt{2})^2$

The triangle is right angled and isosceles.

Ans: (right angled and isosceles.)

(iv) $8 + 10 = 18$

The sum of two sides of a triangle must be greater than the third side.

8, 10 and 18 cannot be the sides of a triangle.

Ans: (triangle cannot be formed.)

9. Let the sides of the triangle be a, b and c . Let $a > b > c$

$a + b + c = 30$

As $a > b > c$, $a + b + c < 3a$

$30 < 3a$

$a > 10$

$b + c > a$

$\therefore a + b + c > 2a$

$30 > 2a$

$a < 15$

$10 < a < 15$

$\therefore a = 11$ or 12 or 13 or 14

If $a = 11$, $b + c = 19$, then (b, c) = (10, 9) only

If $a = 12$, $b + c = 18$, then (b, c) = (11, 7), (10, 8) but not (9, 9)

If $a = 13$, $b + c = 17$, then (b, c) = (12, 5), (11, 6), (10, 7), (9, 8)

If $a = 14$, $b + c = 16$, then (b, c) = (13, 3) (12, 4), (11, 5), (10, 6), (9, 7) but not (8, 8)

Number of values of (a, b, c) = 12

Ans: (12)

10. (i) $AB^2 + BC^2 = 2304 + 196 = 2500 = 50^2 = AC^2$

The triangle ABC is right angled at B.

BD is the median drawn to AC.

D is the midpoint of AC.

As the triangle is right angled at B, D is the circumcenter of triangle ABC.

$BD = AD = CD = \frac{50}{2} = 25$

G is the centroid of triangle ABC

$\therefore$ G lies on the median BD

G divides BD in the ratio 2 : 1 such that BG : GD = 2 : 1

$GD = \frac{1}{3}(BD) = \frac{1}{3}(25) = \frac{25}{3}$ m Ans: ($\frac{25}{3}$) m

- (ii) The medians of a triangle T divide the triangle into six parts of equal area. Hence the area of the triangle formed by the centroid of T and any of the sides of T will be one-third of the area of T.

Area of triangle GQR = $\frac{\text{Area of PQR}}{3} = \frac{30}{3} = 10$ sq units
Ans: (10 sq. Units)

11. (i) Let the length of each equal side be x m
The altitude drawn to the base of the isosceles triangle is also a median of the triangle.

$$x^2 = \left(\frac{28}{2}\right)^2 + (2\sqrt{15})^2 = 196 + 6 = 256$$

$$x = \sqrt{256} = 16$$

Perimeter of the triangle = $2x + 28 = 60$ m

Ans: (60)

- (ii) $XY^2 + YZ^2 = 100 + 576 = 676 = 26^2 = XZ^2$

The triangle XYZ is right angled at Y.

Let the length of the altitude to XZ be h .

$$\text{Area of XYZ} = \frac{1}{2}(XY)(YZ) = \frac{1}{2}(XZ)(h)$$

$$= \frac{1}{2}(10)(24) = \frac{1}{2}(26)(h)$$

$$h = \frac{(10)(24)}{26} = \frac{120}{13}$$
 m Ans: ($\frac{120}{13}$) m

12. (i) The bisectors of angle B and angle C in triangle ABC meet at I.

$$\therefore \angle BIC = 90^\circ + \frac{\angle BAC}{2} = 90^\circ + \frac{48^\circ}{2} = 114^\circ$$

Ans: (114°)

- (ii) The angle bisector of angle P meets QR at S.

$$\therefore \frac{PQ}{PR} = \frac{QS}{SR}$$

$$\frac{QS}{SR} = \frac{PQ}{PR} = \frac{15}{25} = \frac{3}{5}$$

$$QS = \frac{3}{8}QR = \frac{3}{8}(32) = 12$$
 m Ans: (12 m)

13. In triangles RSU and PQU, $\angle RSU = \angle PQU = 90^\circ$

$\angle RUS = \angle PUQ$

The third pair of angles of the two triangles must also be equal

Triangles RSU and PQU are similar

$$\therefore \frac{RS}{PQ} = \frac{SU}{QU} \text{ — (1)}$$

Triangles RSQ and TUQ are similar

$$\therefore \frac{RS}{TU} = \frac{QS}{QU} \text{ — (2)}$$

$$\text{Adding (1) and (2), } \frac{RS}{PQ} + \frac{RS}{TU} = \frac{SU + QS}{QU} = \frac{QU}{QU} = 1$$

$$RS \left(\frac{1}{PQ} + \frac{1}{TU} \right) = 1$$

$$RS \left(\frac{TU + PQ}{(PQ)(TU)} \right) = 1$$

$$RS \frac{(PQ)(TU)}{TU + PQ} = \frac{(36)(45)}{45 + 36} = \frac{(36)(45)}{81} = \frac{4}{9} (45) = 20$$

Ans: (20)

- 14. (i)** Basic proportionality theorem:
Whenever we have points on each of two of the sides of a triangle such that the line joining the two points is parallel to the third side, then the points divide the sides they lie on in the same ratio.

Converse of the basic proportionality theorem:
Whenever we have a point on each of two of the sides of a triangle such that they divide the sides they lie on in the same ratio, then the line joining the points is parallel to the third side.

$$\frac{PA}{AQ} = \frac{PB}{BR} = \frac{4}{3} \text{ — (1)}$$

By converse of basic proportionality theorem, $AB \parallel QR$.

In triangles PAB and PQR, $\angle P$ is common

$\angle PAB = \angle PQR$ ($\because$ corresponding angles are equal)

$\angle PBA = \angle PRQ$ ($\because$ corresponding angles are equal)

Triangle PAB and PQR are similar

$$\frac{PA}{AQ} = \frac{PB}{PR} = \frac{AB}{QR}$$

$$\text{From (1), } \frac{PA}{PQ} = \frac{4}{7}$$

$$\therefore \frac{AB}{QR} = \frac{PA}{PQ} = \frac{4}{7}$$

$$\frac{AB}{70} = \frac{4}{7}$$

$$AB = \frac{4}{7} (70) = 40 \text{ cm} \quad \text{Ans: (40 cm)}$$

- (ii)** Triangles ABC and DEF are similar

$$\frac{BC}{DF} = \frac{AC}{EF} = \frac{AB}{DE}$$

Let each of these equal to k

$$\frac{DF}{EF} = \frac{BC}{AC} = \frac{40}{32} = \frac{5}{4} \text{ — (1)}$$

Perimeter of DEF is 186 cm

$$DE + EF + DF = 186$$

$$60 + EF + DF = 186$$

$$EF + DF = 126 \text{ — (2)}$$

From (1) and (2), $EF = 56$ cm and $DF = 70$ cm

Perimeter of ABC = $AB + BC + AC = k (DE + DF + EF)$

$$= k (186)$$

$$= \frac{AC}{EF} (186) = \frac{32}{56} (186) = \frac{4}{7} (186) = \frac{744}{7} \text{ cm}$$

$$\text{Ans: } \left(\frac{744}{7} \right) \text{ cm}$$

- 15.** In triangle ABD and BCD, $\angle ABD = \angle BDC$

BD is common

$\angle ADB = \angle BDC$

ABD is congruent to CBD

Ratio of the areas of ABD and CBD i.e BCD = 1 : 1

$$\text{Ans: (1 : 1)}$$

ADDITIONAL QUESTIONS FOR PRACTICE

Solutions for questions 1 to 5:

- 1.** Let the third side be x cm.
 $8^2 + 15^2 > x^2$, $8^2 + x^2 > 15^2$ and $15^2 + x^2 > 8^2$
 $289 > x^2$, $x^2 > 161$ and $15^2 + x^2 > 8^2$
 $15^2 + x^2 > 8^2$ is anyway true
 $\therefore 289 > x^2$ and $x^2 > 161$
x is an integer.
 $\therefore x < 17$ and $x \geq 13$ — (1)
Triangle inequality : $15 - 8 < x < 15 + 8$
 $7 < x < 23$ — (2)
From (1) and (2), $13 \leq x < 17$
 $x = 13, 14, 15$ or 16

Ans: (4)

- 2.** SU bisects $\angle PSR$
 $\therefore \frac{PS}{SR} = \frac{PU}{UR} = \frac{128}{16} = 8$
QS : SR = 2 : 1 i.e., QS = 2SR

$$\therefore \frac{PS}{QS} = \frac{PS}{2SR} = \frac{8}{2} = 4$$

ST bisects $\angle PSQ$

$$\therefore \frac{PT}{TQ} = \frac{PS}{QS} = 4$$

$$PT = \frac{4}{5} PQ$$

Given PQ = 100 cm

$$\therefore PT = \frac{4}{5} (100) = 80 \text{ cm}$$

Ans: (80 cm)

- 3.** BF $\parallel$ CE

$$\therefore \frac{AB}{BC} = \frac{AF}{FE} \text{ — (1)}$$

$$\text{CF } \parallel \text{ DE} \\ \therefore \frac{AC}{CD} = \frac{AF}{FE} \text{ — (2)}$$

From (1) and (2),

$$\frac{AB}{BC} = \frac{AC}{CD}$$

$$\frac{AB}{2} = \frac{AB + BC}{3}$$

$$\frac{AB}{2} = \frac{AB + 2}{3}$$

$$AB = 4 \text{ cm}$$

Ans: (4 cm)

4. Let $\angle DFE$ be x°

$$\angle ABC = x^\circ$$

$$DE = EF$$

$$\therefore \angle DEC = \angle EDF + \angle DFE = x^\circ + x^\circ = 2x^\circ$$

$$CD = DE$$

$$\therefore \angle DCE = \angle DEC = 2x^\circ$$

$$\angle CDE = 180^\circ - (\angle DCE + \angle DEC) = 180^\circ - (2x^\circ + 2x^\circ) = (180 - 4x)^\circ$$

$$\angle CDB = 180^\circ - (\angle CDE + \angle EDF) = 180^\circ - ((180 - 4x)^\circ + x^\circ) = 3x^\circ$$

$$BC = CD$$

$$\therefore \angle CBD = \angle CDB = 3x^\circ$$

$$\angle BCD = 180^\circ - (\angle CBD + \angle CDB) = 180^\circ - (3x^\circ + 3x^\circ) = (180 - 6x)^\circ$$

$$\angle ACB = 180^\circ - (\angle BCD + \angle DCE) = 180^\circ - ((180 - 6x)^\circ + 2x^\circ) = 4x^\circ$$

$$\angle BAC = 180^\circ - (\angle ABC + \angle ACB) = 180^\circ - (x^\circ + 4x^\circ) = (180 - 5x)^\circ$$

$$AB = BC$$

$$\therefore \angle BAC = \angle BCA$$

$$(180 - 5x)^\circ = 4x^\circ$$

$$180^\circ - 5x^\circ = 4x^\circ$$

$$9x = 180^\circ$$

$$x = 20$$

$$\angle DFE = x^\circ = 20^\circ$$

Ans: (20°)

5. AD is the angular bisector of $\angle BAC$

$$\therefore \frac{AB}{AC} = \frac{BD}{DC}$$

$$\frac{2BD}{AC} = \frac{BD}{DC}$$

$$AC = 2DC$$

$$AC = BC$$

$$2DC = BD + DC$$

$$DC = BD$$

$$2DC = 2BD$$

$$AC = AB$$

$$AC = AB = BC \text{ i.e. triangle ABC is equilateral ——— (1)}$$

The angular bisectors of the angles of ABC meet at I.

$\therefore$ I is the incentre of ABC

$$\therefore \angle BIC = 90^\circ + \frac{\angle BAC}{2} = 90^\circ + \frac{60^\circ}{2} = 120^\circ (\therefore \text{From (1)})$$

Ans: (120°)